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Date: 12/01/25

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Course: ECON 1BB3: Introductory
Macroeconomics

Assignment: Chapter 10

1. A baseball fan with a Shohei Ohtani baseball card wants to trade it for an Aaron Judge baseball card, but everyone the fan knows who has a Judge card doesn't want an Ohtani card.

Economists characterize this problem as a failure of the

- ☒ A. principle of a double coincidence of wants.
- ☐ B. irrational exuberance doctrine.
- ☐ C. theory of comparative advantage.
- ☐ D. market clearing mechanism.

2. Which of the following best explains the difference between commodity money and fiat money?

- ☐ A. Commodity money has no value except as money, whereas fiat money has value independent of its use as money.
- ☒ B. Fiat money has no value except as money, whereas commodity money has value independent of its use as money.
- ☐ C. All money is commodity money, as it has to be exchanged for gold by the central bank.
- ☐ D. Commodity money is usually authorized by the central bank, whereas fiat money has to be exchanged for gold by the central bank.

3. Which one of the following is not a function of money?

- ☐ A. Store of value.
- ☒ B. Open market operation.
- ☐ C. Unit of account.
- ☐ D. Medium of exchange.

If something is to be considered as money, it has to fulfill all four functions .

4. The economic definition of money is:

- ☐ A. Anything of value owned by a person or a firm.
- ☒ B. Any asset that people are generally willing to accept in exchange for goods and services.
- ☐ C. A good that has intrinsic value.
- ☐ D. Anything authorized by the government to be used in an exchange.

5. When money is acting as a store of value, it allows an individual to

- ☒ A. transfer dollars, and therefore purchasing power, into the future.
- ☐ B. measure the value of goods and services in the economy.
- ☐ C. trade money for goods and services in the economy.
- ☐ D. exchange goods for other goods and services in the economy.

Money is an imperfect standard of deferred payment because inflation causes the value of money to decrease over time.

6. An asset would be usable as a medium of exchange for all of the following reasons except:

- ☐ A. the asset must be generally accepted by most people.
- ☐ B. the asset should be durable and not lose value due to spoilage.
- ☒ C. the asset should be a commodity that has intrinsic value.
- ☐ D. the asset should be divisible since goods are valued at different amounts.

Which of the following would be the least desirable candidate to be a good medium of exchange?

- ☐ A. dollar bills
 - ☐ B. seashells
 - ☐ C. gold
 - ☒ D. milk
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7. The use of money

- ☐ A. eliminates the double coincidence of wants.
 - ☐ B. allows for greater specialization.
 - ☐ C. reduces the transaction costs of exchange.
 - ☒ D. all of the above.
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8. What are the largest asset and the largest liability of a typical bank?

- ☐ A. Reserves are the largest asset and deposits are the largest liability of a typical bank.
 - ☐ B. Cash in its vault is the largest asset and bonds are the largest liability of a typical bank.
 - ☐ C. Loans are the largest liability and deposits are the largest asset of a typical bank.
 - ☒ D. Loans are the largest asset and deposits are the largest liability of a typical bank.
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9. Suppose you decide to withdraw \$100 in cash from your checking account. Which one of the following choices accurately shows the effect of this transaction on your bank's balance sheet.

- ☒ A. Your bank's balance sheet shows a decrease in reserves by \$100 and a decrease in deposits by \$100.
 - ☐ B. Your bank's balance sheet shows a decrease in reserves by \$100 and an increase in deposits by \$100.
 - ☐ C. Your bank's balance sheet shows an increase in reserves by \$100 and an increase in deposits by \$100.
 - ☐ D. Your bank's balance sheet shows an increase in reserves by \$100 and a decrease in deposits by \$100.
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10.

Suppose you deposit \$3500 in currency into your chequing account at a branch of Bank X, which we will assume has no excess reserves¹ at the time you make your deposit. Also assume that the desired reserve ratio² is 0.25, or 25 percent.

a. Use the first row of the first T-account to the right to show the initial impact of this transaction on Bank X's balance sheet.

b. Suppose that Bank X makes the maximum loan it can from the funds you deposited. Using the second and third rows of the first T-account to the right, show the initial impact of granting the loan on Bank X's balance sheet. Assume the loan is made through the borrower's chequing account.

c. Now suppose that whoever took out the loan in (b) writes a cheque for this amount and that the person receiving the cheque deposits it in a branch of Bank Y. Show the effect of these transactions on the balance sheets of Bank X and Bank Y after the cheque has been cleared. (On the T-account for Bank X, show the net effect of all transactions, including the transactions from parts (a) and (b).)

d. What is the maximum increase in chequing account deposits that can result from your \$3500 deposit?

The maximum increase in chequing account deposits is

\$.

What is the maximum increase in the money supply? Explain.

Select the correct choice below and fill in the answer box to complete your choice.

- ☐ A. The maximum increase in the money supply is \$ because the money supply changes by the increase in chequing account deposits.
- ☐ B. The maximum increase in the money supply is \$ because the money supply changes by the increase in chequing account deposits plus money already counted as part of the supply.
- ☒ C. The maximum increase in the money supply is \$ because the money supply changes by the increase in chequing account deposits minus money already counted as part of the supply.
- ☐ D. The maximum increase in the money supply is \$ because the money supply changes by the amount of the original deposit.

Bank X			
Assets		Liabilities	
(a)	Reserves + \$3500	Deposits + \$3500	
(b)	Reserves + \$0	Deposits + \$2625	
	Loans + \$2625		

(c) Bank X			
Assets		Liabilities	
	Reserves + \$875	Deposits + \$3500	
	Loans + \$2625		

(c) Bank Y			
Assets		Liabilities	
	Reserves + \$2625	Deposits + \$2625	
	Loans + \$0		

1: Definition

Excess reserves are reserves that banks hold over and above the desired amounts.

2: Definition

The minimum fraction of deposits that banks desire to keep as reserves is called the desired reserve ratio.

11. Briefly explain whether you agree or disagree with the following statement:

"Assets³ are things of value that people own. Liabilities are debts. Therefore, a bank will always consider a checking account deposit to be an asset and a car loan to be a liability."

- ☐ A. Agree. Loans are debts and therefore bank liabilities.
- ☒ B. Disagree. Checking accounts represent something that the bank owes to the owner of the account. It is a bank liability.
- ☐ C. Agree. Checking accounts are something of value that is in the bank. Therefore, they are bank assets.
- ☐ D. Disagree. Both checking accounts and car loans represent bank liabilities.

3: More Info

Asset is anything of value owned by a person or a firm.

12. Suppose banks keep no excess reserves and that all banks are currently meeting their desired reserve requirement. The Bank of Canada then makes an open market purchase of \$10,000 from Bank 1.

Use the T-account below to show the result of this transaction for Bank 1, assuming Bank 1 keeps no excess reserves after the transaction. (Remember T-accounts show the changes to a bank's balance sheet.)

Bank 1's T-account	
Assets	Liabilities
Reserves	Deposits
\$ 0	\$ 0
Loans	
\$ + 10000	
Securities	
\$ - 10000	

Assume all of Bank 1's loans of \$10,000 are spent by the borrowers and then deposited into Bank 2.

Use the T-account below to show the result of this transaction for Bank 2, assuming Bank 2 keeps no excess reserves and the desired reserve ratio is 14%. (Remember T-accounts show the changes to a bank's balance sheet.)

Bank 2's T-account	
Assets	Liabilities
Reserves	Deposits
\$ + 1400	\$ + 10000
Loans	
\$ + 8600	

Assume all of Bank 2's loans of \$8,600 are spent by the borrowers and then deposited into Bank 3.

Use the T-account below to show the result of this transaction for Bank 3, assuming Bank 3 keeps no excess reserves and the desired reserve ratio is 14%. (Remember T-accounts show the changes to a bank's balance sheet.)

Bank 3's T-account	
Assets	Liabilities
Reserves	Deposits
\$ + 1204	\$ + 8600
Loans	
\$ + 7396	

13. Why does an open market purchase of government securities by the Bank of Canada increase bank reserves? Why does an open market sale of government securities by the Bank of Canada decrease bank reserves?

When the Bank of Canada purchases government securities in the open market,

- ☐ A. the sellers of such securities buy new securities in the open market and there is an increase in bank reserves.
- ☒ B. the sellers of such securities deposit the funds in their banks and bank reserves increase.
- ☐ C. the buyers of these securities pay for them with checks drawn on their bank account and bank reserves increase.
- ☐ D. the public starts buying houses and firms invest in anticipation of bank increasing their reserves.

When the Bank of Canada sells government securities in the open market,

- ☐ A. the public starts selling houses and firms disinvest in anticipation of banks decreasing their reserves.
- ☒ B. the buyers of these securities pay for them with checks and bank reserves fall.
- ☐ C. the buyers of such securities buy new securities in the open market and there is a decrease in bank reserves.
- ☐ D. the sellers of such securities deposit the funds in their banks and bank reserves decrease.

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14. Which one of the following is not the formula for the quantity theory of money?

- ☐ A. $V = \frac{P \times Y}{M}$.
- ☐ B. $M \times V = P \times Y$.
- ☐ C. $M = \frac{1}{V} \times P \times Y$.
- ☒ D. $M \times Y = P \times V$.

How does the quantity theory provide an explanation about the cause of inflation?

- ☐ A. The quantity equation shows that if the money supply grows at a faster rate than nominal GDP, then there will be inflation.
- ☒ B. The quantity equation shows that if the money supply grows at a faster rate than real GDP, then there will be inflation.
- ☐ C. The quantity equation shows that if the money supply grows at a slower rate than real GDP, then there will be inflation.
- ☐ D. The quantity equation shows that if the money supply grows at the same rate as real GDP, then there will be inflation.

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15. Is the quantity theory of money⁴ better able to explain the inflation rate in the long run or in the short run? Briefly explain.

The quantity theory of money is better able to explain the inflation rate in the long run because in the short run there can be significant fluctuations in velocity.

4: Definition

The quantity theory of money is a theory about the connection between money and prices that assumes that the velocity of money is constant.

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16. If the money supply is growing at a rate of 9 percent per year, real GDP (real output) is growing at a rate of 4 percent per year, and velocity⁵ is constant, what will the inflation rate be?

% (Round your response to the nearest whole number.)

If the money supply is growing at a rate of 9 percent per year, real GDP (real output) is growing at a rate of 4 percent per year, and velocity is growing at 2 percent per year instead of remaining constant, what will the inflation rate be?

% (Round your response to the nearest whole number.)

5: More Info

Velocity of money is the average number of times each dollar in the money supply is used to purchase goods and services included in GDP.

17. Which of the following is true with respect to *hyperinflation*?

- ☐ A. It is caused by central banks increasing the money supply at a rate much greater than the growth rate of real GDP.
- ☐ B. It can be hundreds—even thousands—of percentage points per year.
- ☐ C. In the presence of hyperinflation, firms and households avoid holding money.
- ☒ D. All of the above.

18. Using the following information what is the velocity of money?

Component	Value
Money supply	\$1,100
Price level	1.98
Real GDP	\$8,000

The velocity of money is equal to: (enter your answer rounded to two decimal places).

19. Arkensland is a highly industrialized country. Prices have been steadily increasing over the last few years and inflation reached an all-time high of 8 percent last year. Following this, the central bank came under pressure to reduce inflation to at least 4 percent in this year. With real GDP expected to grow at 2 percent this year, the finance minister suggested that the central bank should ensure that money supply growth in the current year is not more than 6 percent. However, inflation at the end of this year turned out to be higher than 4 percent, even though real GDP grew at 2 percent and money supply growth was restricted to 6 percent.

Which of the following can most reasonably be inferred from the above information?

- ☐ A. There is a risk of hyperinflation in Arkensland.
- ☒ B. The velocity of money increased this year.
- ☐ C. The central bank of Arkensland must have sold bonds this year in order to reduce the money supply.
- ☐ D. The growth of GDP in Arkensland has stagnated.
- ☐ E. Even though the inflation target was not met, the overall price level did decline this year as compared to last year.

